Type MS · Willingham's learning theory

Answer 1 of 9 · Question 4 part b · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number	Answer	Mark
4(a)	**One** mark for each conclusion made (maximum **two** marks). **One** mark for justification of each conclusion through analysis/interpretation (maximum **two** marks). For example: • The children in the class have not made much improvement over the four days (1) with an increase of only 1% in the average percentage getting the order of planets correct from day 1 to day 4 assessments (1). • The children have not got quicker on the task on average over the four days (1) which is shown by the average time taken to name the planets in order from the Sun remaining at around 21 seconds from day 1 to day 4 assessments (1). Accept any other appropriate response.	(4)

Question number	Answer	Mark
4(b)	**One** mark for accurate understanding of Willingham's learning theory. **One** mark for exemplification of how the theory can be used to explain the scenario. For example: • Willingham's learning theory suggests that in order to improve a new cognitive skill, practice is required using effort (1) so Mr Harris could ask his class to put pictures of the planets in the correct order repeatedly to reinforce the correct order of planets which will help them practise the new skill (1). Accept any other appropriate response.	(2)

Type MS · Willingham's learning theory

Answer 1 of 9 · Question 4 part b · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Section B
Memory – How does your memory work?

Type MS · Willingham's learning theory

Answer 2 of 9 · Question 3 part a · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for accurate understanding of Willingham's learning theory. **One** mark for exemplification of how the theory can be used to explain the scenario. For example: • Willingham's theory suggests that children need to practise movement skills repeatedly to make them automatic (1). Mario could involve the pupils in activities such as cutting using scissors repeatedly, offering corrective feedback (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
3(b)	**One** mark for identification of each strength or weakness (maximum **two** marks). **One** mark for justification which must be linked to the strength or weakness identified (maximum **two** marks). For example: Strength • Willingham's theory has received experimental support which means practising the skills in Mario's class could help (1), such as that of Ericsson, Krampe, Tesch-Romer (1993) who found the best violinists engaged in more practice during their training and supports using practice for cutting skills in Mario's class (1). Weakness • The theory does not account for individual differences in the physical development of Mario's pupils (1), which means that some pupils in Mario's class may not have their needs met by the universal strategies suggested by Willingham and could need more individualised support (1). Accept any other appropriate response.	(4)

Type MS · Willingham's learning theory

Answer 2 of 9 · Question 3 part a · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Type MS · Willingham's learning theory

Answer 3 of 9 · Question 3 part b · 4 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for accurate understanding of Willingham's learning theory. **One** mark for exemplification of how the theory can be used to explain the scenario. For example: • Willingham's theory suggests that children need to practise movement skills repeatedly to make them automatic (1). Mario could involve the pupils in activities such as cutting using scissors repeatedly, offering corrective feedback (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
3(b)	**One** mark for identification of each strength or weakness (maximum **two** marks). **One** mark for justification which must be linked to the strength or weakness identified (maximum **two** marks). For example: Strength • Willingham's theory has received experimental support which means practising the skills in Mario's class could help (1), such as that of Ericsson, Krampe, Tesch-Romer (1993) who found the best violinists engaged in more practice during their training and supports using practice for cutting skills in Mario's class (1). Weakness • The theory does not account for individual differences in the physical development of Mario's pupils (1), which means that some pupils in Mario's class may not have their needs met by the universal strategies suggested by Willingham and could need more individualised support (1). Accept any other appropriate response.	(4)

Type MS · Willingham's learning theory

Answer 3 of 9 · Question 3 part b · 4 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Type MS · Willingham's learning theory

Answer 4 of 9 · Question 1 part a · 1 mark

2021 NOV · Paper 01 · find part a in this mark scheme extract

Question number	Answer	Mark
1(a)	**D** - Factual knowledge precedes skill The only correct answer is D. A is incorrect because this is not a concept in the theory. B is incorrect because this is not a concept in the theory. C is incorrect because this is not a concept in the theory.	(1)

Question number	Answer	Mark
1(b)	**B** - Practice and effort The only correct answer is B. A is incorrect because this was not emphasised in his theory. C is incorrect because this was not emphasised in his theory. D is incorrect because this was not emphasised in his theory.	(1)

Type MS · Willingham's learning theory

Answer 5 of 9 · Question 1 part b · 1 mark

2021 NOV · Paper 01 · find part b in this mark scheme extract

Question number	Answer	Mark
1(a)	**D** - Factual knowledge precedes skill The only correct answer is D. A is incorrect because this is not a concept in the theory. B is incorrect because this is not a concept in the theory. C is incorrect because this is not a concept in the theory.	(1)

Question number	Answer	Mark
1(b)	**B** - Practice and effort The only correct answer is B. A is incorrect because this was not emphasised in his theory. C is incorrect because this was not emphasised in his theory. D is incorrect because this was not emphasised in his theory.	(1)

Type MS · Willingham's learning theory

Answer 6 of 9 · Question 3 part whole · 2 marks

2021 NOV · Paper 01 · find part whole in this mark scheme extract

Question number	Answer	Mark
3	**One** mark for demonstrating accurate understanding of Daniel Willingham's learning theory. **One** mark for exemplification of how the theory can be used to explain the scenario. For example: • Willingham's theory would advocate the use of demonstrating the appropriate social behaviour by others, such as the teacher or peers, to improve social development (1), so Mrs Kumar could create activities where Molly and Mack work with children that do not snatch and ask nicely when wanting to play with a toy, so that their behaviour could be modelled by Molly and Mack (1). Accept any other appropriate response.	**(2)**

Type MS · Willingham's learning theory

Answer 7 of 9 · Question 5 part a · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
5(a)	**One** mark for appropriate conclusion. **One** mark for justification of the conclusion through analysis/interpretation. For example: • Social learning appears to have improved the social development and specifically the sharing behaviour of the children (1) which is shown by the higher score given to all of those supervised by James (all scored 7-9) who used social learning compared to those supervised by the other teacher (all scored 4-6) (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
5(b)	**One** mark for identification of a relevant improvement. **One** mark for justification, which must be linked to the improvement identified. For example: • James could have asked someone independent to judge the ability of the children to share and take turns because he may have been biased (1) whereas another person would not know whether the children were exposed to social learning or not so would have made objective decisions regarding the ability to share and take turns (1). Accept any other appropriate response.	(2)

Type MS · Willingham's learning theory

Answer 7 of 9 · Question 5 part a · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Section B
Memory – How does your memory work?

Type MS · Willingham's learning theory

Answer 8 of 9 · Question 4 part a · 4 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
4(a)	**One** mark for each appropriate conclusion (maximum **two** marks). **One** mark for justification of each conclusion through analysis/interpretation (maximum **two** marks). For example: • One conclusion is that practice and effort improves juggling skills (1), which is shown by Sergio's niece being able to juggle for 10 seconds longer by day 7 (14 seconds) compared to day 1 (4 seconds) (1). • A conclusion could be that getting supportive messages praising effort helps to encourage practise and learning to juggle (1) which is shown by his niece being able to juggle for 10 seconds longer than his nephew by day 7, who did not get any messages (1). Accept any other appropriate response.	**(4)**

Question number	Answer	Mark
4(b)	**One** mark for identification of each weakness (maximum **two** marks). **One** mark for justification which must be linked to the weakness identified (maximum **two** marks). For example: Weakness one • Sergio only used his own niece and nephew so the investigation lacks generalisability as they may not reflect others when juggling (1) as his niece may have been naturally gifted at juggling and her improvement may not have been shown by other individuals within the same time period (1). Weakness two • Sergio did not specify exactly how long his niece should practise for or check that his nephew did not practise juggling (1) so the lack of controls used in his investigation means that it is difficult to infer cause and effect as the practice and praise may not have actually caused his niece's improvement (1). Accept any other appropriate response.	**(4)**

Type MS · Willingham's learning theory

Answer 8 of 9 · Question 4 part a · 4 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Section B
Memory – How does your memory work?

Type MS · Willingham's learning theory

Answer 9 of 9 · Question 3 part a · 4 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for each appropriate conclusion (maximum **two** marks). **One** mark for justification of each conclusion through analysis/interpretation (maximum **two** marks). For example: • Doing a task that is more challenging and reasonably difficult leads to the most effective learning (1) which is shown by the group doing the reasonably difficult task improving by 30% which is the most of all groups (1). • Doing a task that is too difficult can lead to no improvement in learning a new concept (1) which is shown by the group doing the very complex task having no improvement to still achieving 50% by the end of the week (1). • Doing a very easy task gives less improvement in learning a new concept than a more challenging, reasonably difficult task (1) which is shown by the group doing the very easy task showing a 10% improvement by the end of the week compared a 30% improvement by those doing a more challenging, reasonably difficult task (1). Accept any other appropriate response.	**(4)**

Type MS · Willingham's learning theory

Answer 9 of 9 · Question 3 part a · 4 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Question number	Answer	Mark
3(b)	**One** mark for identification of a strength/weakness (maximum **two** marks). **One** mark for justification which must be linked to the strength/weakness identified (maximum **two** marks). For example: Strength • Alessia gave the students a test at the beginning of the week which gave them a baseline measure of performance on the new science concept (1) which enabled Alessia to accurately measure how effective the task given was as she could compare before to after with the same students which increases validity (1). • Alessia gave all the students in each group the same task with the same activities included so there were some controls (1) which is a strength because Alessia can test her findings about difficulty of task when learning a new concept for reliability to see if she gets consistent results from her students about learning. Weakness • Alessia gave the different tasks to her students in groups during their lessons at school so they could have been distracted or helped by others (1) which means that Alessia cannot be sure it was the task that led to the test performance at the end of the week as being helped by other students could have improved their understanding of the science concept (1). • Alessia only recruited her own students from a single school to investigate difficulty of task in learning a new concept so this lacks generalisability to all students (1) because her students may all have a high prior attainment so any conclusions about difficulty of task in learning something new would be affected by this and so not be representative of students in general (1). Accept any other appropriate response.	**(4)**

Type MS · Willingham's learning theory

Answer 9 of 9 · Question 3 part a · 4 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract